5.3 Principles of Electrochemistry (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.3 Principles of Electrochemistry (A Level Only)
Difficulty	Medium

Time allowed: 70

Score: /57

Percentage: /100

Question 1a

This question is about electrochemical cells.

Fig. 1.1 shows the apparatus used to measure the standard electrode potential, E^{θ}, of a cell composed of a Cu(II)/Cu half-cell and an Fe(II)/Fe half-cell.

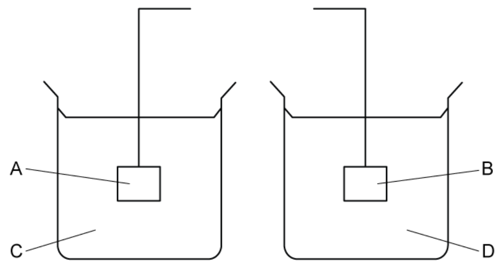


Fig. 1.1

Finish the diagram by adding components to show the complete circuit. Label the components you add.

[2 marks]

Question 1b

In the spaces below, identify or describe what the four letters **A**-**D** in **Fig. 1.1** represent.

A ....

B ....

C ....

D ....

[3 marks]

Question 1c

The standard electrode potentials of the two half-cells are as follows.

$$Fe^{2+} + 2e^- \rightleftharpoons Fe \qquad E^\theta = -0.44\ V$$
$$Cu^{2+} + 2e^- \rightleftharpoons Cu \qquad E^\theta = +0.34\ V$$

i)
Identify which electrode acts as the negative electrode.

[1]

ii)
Write the overall equation of the electrochemical cell.

[2]

[1 mark]

Question 1d

E^θ_{cell} of this cell is +0.78 V.

State whether this reaction is feasible. Explain your answer.

[1 mark]

Question 2a

A student decided to determine the value of the Faraday constant by an electrolysis experiment. Fig. 2.1 is an incomplete diagram showing the apparatus that was used.

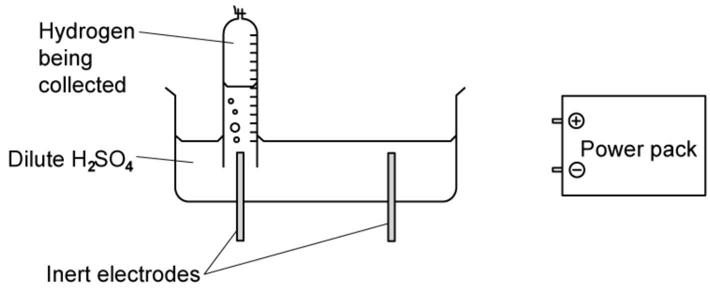


Fig. 2.1

i)

Apart from connecting wires, what **two** additional pieces of equipment are needed for this experiment?

[2]

ii)

Complete the diagram, showing additional equipment connected in the circuit, and showing the powerpack connected to the correct electrodes.

[2]

[4 marks]

Question 2b

List the measurements the student would need to make in order to use the results to calculate a value for the Faraday constant.

[3 marks]

Question 2c

Using an equation, state the relationship between the Faraday constant, *F*, the Avogadro constant, *L*, and the charge on the electron, *e*.

[1 mark]

Question 2d

The value the student obtained was: 1 faraday = 9.65×10^4 coulombs

Use this value and your equation in **(c)** to calculate the Avogadro constant (take the charge on the electron to be 1.60×10^{-19} coulombs).

[1 mark]

Question 3a

This question is about electrochemical cells.

The diagram shown in **Fig. 3.1** represents the standard hydrogen electrode.

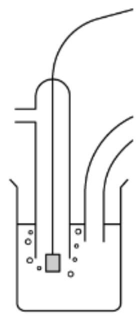


Fig 3.1

i)
Name the substance used as the electrode in **Fig 3.1**.

[1]

ii)
Suggest why this substance is used as an electrode.

[2]

iii)
Give the standard conditions used in a standard hydrogen electrode.

[3]

[6 marks]

Question 3b

A student set up an electrochemical cell consisting of copper and zinc.

i)

Complete **Fig. 3.2** to show the components and reagents, including their concentrations and label any apparatus required to complete the electrochemical cell.

[4]

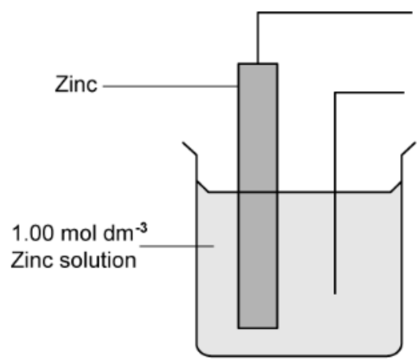


Fig. 3.2

ii)

Use the IUPAC convention to give the half-equations occurring at each electrode.

[2]

[6 marks]

Question 3c

A student set up another electrochemical cell consisting of copper and silver as shown in the following cell representation where the half cell with the greatest negative standard electrode potential is written on the left:

$$Cu / Cu^{2+} ; Ag^{+} / Ag$$

i)
Write a half-equation for the reaction that occurs at the positive electrode.

[1]

ii)
Write a half-equation for the reaction that occurs at the negative electrode.

[1]

iii)
Use the half-equations to deduce an overall equation for the cell. Include all state symbols.

[1]

[3 marks]

Question 3d

A diagram of a cell is shown below in **Fig. 3.3**.

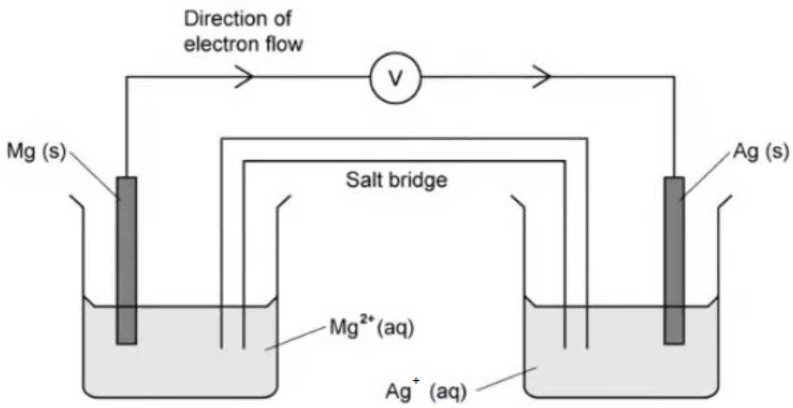


Fig. 3.3

i)

Explain how the salt bridge, in **Fig. 3.3**, provides an electrical connection between the two solutions.

[1]

ii)

Suggest why potassium chloride would **not** be suitable for use in the salt bridge of this cell.

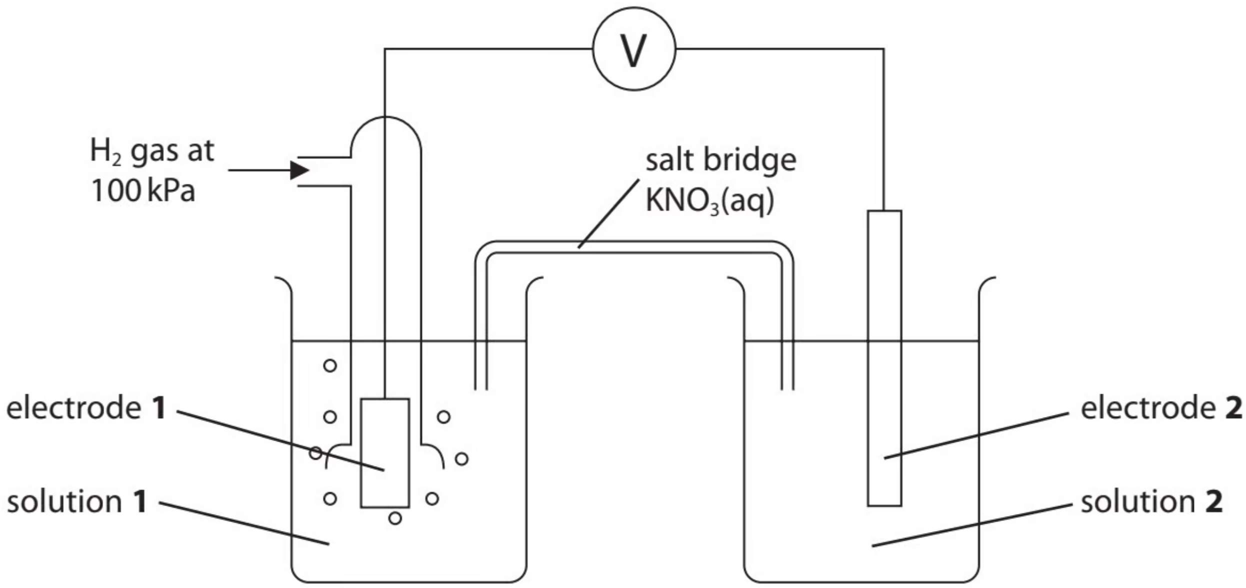
[1]

[2 marks]

Question 4a

The apparatus shown was used to measure the standard electrode potential for the reduction of $Cr_2O_7^{2-}$ ions to Cr^{3+} ions in acid solution:

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$



Which material should be used for each electrode?

Electrode 1 ....

Electrode 2 ....

[2 marks]

Question 4b

Suggest a suitable solution that could be used as solution **1**.

[1 mark]

Question 4c

Solution **2** contains 14.71g of $K_2Cr_2O_7$.

What mass of $Cr_2(SO_4)_3 \cdot 18H_2O$ should also be used?

[M_r values: $K_2Cr_2O_7$ = 294.2 $Cr_2(SO_4)_3 \cdot 18H_2O$ = 716.3]

[3 marks]

Question 4d

Solution **2** is best acidified with H_2SO_4 instead of HCl or HBr.

Suggest why.

[2 marks]

Question 5a

The electrolytic purification of copper can be carried out in an apparatus similar to the one shown in **Fig. 5.1**.

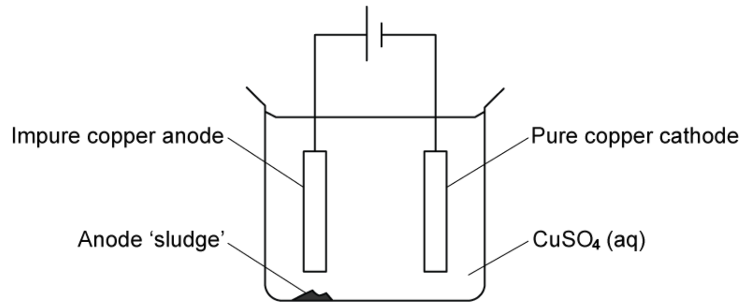


Fig. 5.1

The impure copper anode contains small quantities of metallic nickel, zinc and silver, together with inert oxides and carbon resulting from the initial reduction of the copper ore with coke. The copper goes into solution at the anode, but the silver remains as the metal and falls to the bottom as part of the anode 'sludge'. The zinc also dissolves.

Table 5.1 shows a list of standard electrode potentials at 298 K.

Table 5.1

Electrode reaction	E^θ **/ V**
$Ag^+ + e^- \rightleftharpoons Ag$	+0.80
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+0.34
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	-0.44
$Ni^{2+} + 2e^- \rightleftharpoons Ni$	-0.25
$SO_4^{2-} + 4H^+ + 2e^- \rightleftharpoons SO_2 + 2H_2O$	+0.17
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	-0.76

i)

Write a half-equation including state symbols for the reaction of copper at the anode.

[1]

ii)

Use data from **Table 5.1** to explain why silver remains as the metal.

[2]

iii)

Use data from **Table 5.1** to predict what happens to the nickel at the anode.

[2]

iv)

Write a half-equation including state symbols for the main reaction at the cathode.

[1]

v)

Use data from **Table 5.1** to explain why zinc is not deposited on the cathode.

[1]

[7 marks]

Question 5b

As the electrolysis proceeds, the blue colour of the electrolyte slowly fades.

Suggest why the blue colour fades.

[2 marks]

Question 5c

Most of the current passed through the cell is used to dissolve the copper at the anode and precipitate pure copper onto the cathode. However, a small proportion of it is 'wasted' in dissolving the impurities at the anode which then remain in solution. When a current of 20.0 A was passed through the cell for 10.0 hours, it was found that 225 g of pure copper was deposited on the cathode.

[Faraday constant, $F = 9.65 \times 10^4$ C mol^{-1}]

i)
Calculate the number of moles of copper produced at the cathode.

[1]

ii)
Calculate the number of moles of electrons needed to produce this copper.

[1]

iii)
Calculate the number of moles of electrons that passed through the cell.

[2]

iv)
Hence calculate the percentage of the current through the cell that has been 'wasted' in dissolving the impurities at the anode.

[1]

[5 marks]

Question 5d

Nickel often occurs in ores along with iron. After the initial reduction of the ore with coke, a nickel-iron alloy is formed.

Use data from **Table 5.1** to explain why nickel can be purified by a similar electrolysis technique to that used for copper, using an impure nickel anode, a pure nickel cathode, and nickel sulfate as the electrolyte.

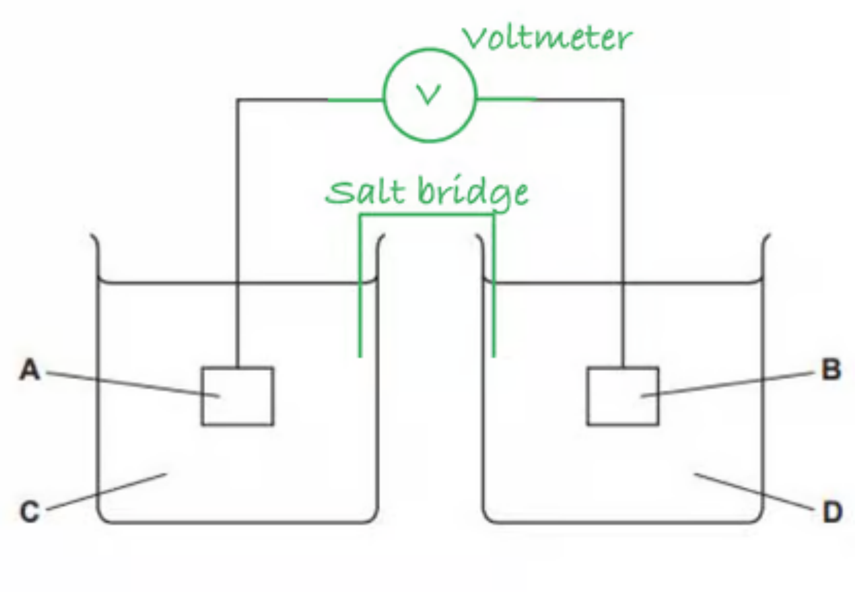
Explain what would happen to the iron during this process.

[2 marks]

Model Answers: Medium

1a

a) The labelled components to show the complete circuit are:



- Voltemeter or V; [1 mark]
- Salt bridge; [1 mark]

[Total: 2 marks]

- A voltmeter is required to measure the standard electrode potential of the cell
  - The wires and voltmeter make up the external part of the circuit through which **electrons** can flow
  - Ensure that your diagram does not have any gaps in the external circuits make sure that you extend the wires to touch the voltmeter
  - If there are any gaps, the circuit is not complete
- The salt bridge connects the two solutions, so it must be drawn so that it is in both solutions
  - The salt bridge allows **ions** to flow, without it, the circuit would be incomplete
  - Typically, the salt bridge is a strip of filter paper that has been soaked in a saturated solution of potassium nitrate

1b

b) The four letters represent:

- **A** and **B**: copper (metal) / Cu **AND** iron (metal) / Fe; [1 mark]
- **C**: 1 mol dm^{-3}
  OR
  D: 1 mol dm^{-3}; [1 mark]
- **C** and **D**: Cu^{2+} / $CuSO_4$ / $CuCl_2$ / $Cu(NO_3)_3$, etc. **AND** Fe^{2+} / $FeSO_4$, etc.; [1 mark]

[Total: 3 marks]

- The half-cells are both metal / metal ion half-cells so the electrode is made from the metal and the solution contains an aqueous solution of the metal ion
- As the standard electrode potential is to be measured, the conditions need to be standard and the concentration of any solutions must be 1 mol dm^{-3}
- The Fe(II)/Fe half-cell must contain a solution with Fe^{2+} ions in, so any appropriate solution containing Fe^{2+} ions is accepted
- Similarly, for the Cu(II)/Cu half-cell, any appropriate solution containing the Cu^{2+} is accepted

1c

c)

i) The electrode that acts as the negative electrode is:

- Iron (metal) / Fe; [1 mark]

ii) The overall equation of the electrochemical cell is:

- $Cu^{2+} + Fe \rightarrow Cu + Fe^{2+}$; [1 mark]

[Total: 2 marks]

- The half-cell with the more negative standard electrode potential will lose electrons more readily making it the negative pole
- Electrons flow from the positive pole to the negative pole so from Cu(II)/Cu half-cell to the Fe(II)/Fe half-cell
- The two half equations are:

- $Cu^{2+} + 2e^- \rightarrow Cu$
- $Fe \rightarrow Fe^{2+} + 2e^-$

- To construct the overall equation, combine the two half equations, there are already the same number of electrons on either side, so when the half-equations are combined, these will cancel out:
  - $Cu^{2+} + \cancel{2e^-} + Fe \rightarrow Cu + Fe^{2+} + \cancel{2e^-}$
  - $Cu^{2+} + Fe \rightarrow Cu + Fe^{2+}$

1d

d) To explain whether this reaction is feasible:

- (The reaction is) feasible; [1 mark]
- (Because) E^{θ}_{cell} is positive; [1 mark]

[Total: 2 marks]

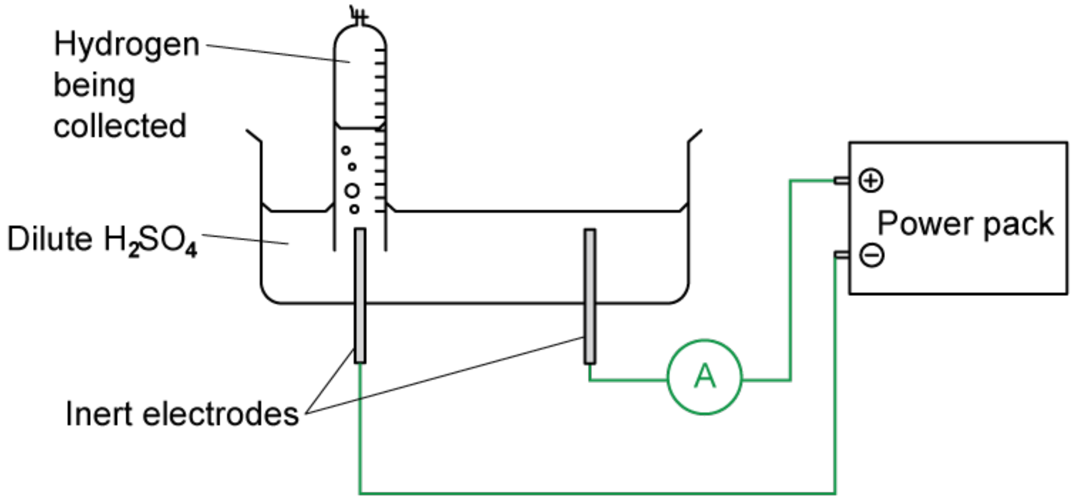
- A reaction is likely to occur when E^{θ}_{cell} is positive

2a

a) The **two** additional pieces of equipment that are needed for this experiment are:

- Ammeter / galvanometer; [1 mark]
- Clock / watch / timer
  OR
  Variable resistor / rheostat; [1 mark]

ii) The completed diagram is:



- Diagram to show ammeter in circuit
  AND
  A complete circuit; [1 mark]
- The negative terminal connected to the left hand electrode; [1 mark]

[Total: 4 marks]

- The current and time are needed to be measured so that the charge can be calculated
- A variable resistor is an acceptable alternative answer
- You must show the ammeter in the circuit with the wires connected correctly so that the circuit is complete, the ammeter can be shown anywhere in the circuit
- Hydrogen gas will collect at the cathode as hydrogen ions will gain electrons:
  - $2H^+ + 2e^- \rightarrow H_2$
- So the electrode on the left hand side must be connected to the negative terminal on the power supply and the other electrode must be connected to the positive terminal
- Oxygen gas will collect at the anode:
  - $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

2b

b) The measurements the student would need to make in order to use the results to calculate a value for the Faraday constant are:

- Volume / amount of hydrogen gas; [1 mark]
- Time; [1 mark]

- Current / amps / ammeter reading; [1 mark]

[Total: 3 marks]

- Often experiments are completed to calculate Avogardo's constant, however, this is an experiment to calculate the Faraday constant
- One faraday is the amount of electric charge carried by 1 mole of electrons or 1 mole of singly charged ions
- So, to calculate the Faraday constant, you need to divide the charge that passes through the electrolyte by the number of moles of electrons that carried this charge
- To find the charge that passes through the electrolyte:
  - Q = *It*
  - So, you need to know the **current** and the **time**
- You also need to find the number of electrons required to produce the amount of hydrogen gas collected
- To calculate this, you need to know how many moles of hydrogen gas were collected from the **volume of gas collected**
- 2 moles of electrons are required to produce 1 mole of hydrogen:
  - $2H^+ + 2e^- \rightarrow H_2$
- So, moles of electrons = 2 x moles of hydrogen collected

2c

c) The relationship between the Faraday constant, *F*, the Avogadro constant, *L*, and the charge on the electron, *e*, is:

- *F* = Le
  OR
  L = *F*/ *e*
  OR

- To find the charge that passes through the electrolyte:
  - $Q = It$
  - So, you need to know the **current** and the **time**
- You also need to find the number of electrons required to produce the amount of hydrogen gas collected
- To calculate this, you need to know how many moles of hydrogen gas were collected from the **volume of gas collected**
- 2 moles of electrons are required to produce 1 mole of hydrogen:
  - $2H^+ + 2e^- \rightarrow H_2$
- So, moles of electrons = 2 x moles of hydrogen collected

2c

c) The relationship between the Faraday constant, *F*, the Avogadro constant, *L*, and the charge on the electron, *e*, is:

- F = Le
  OR
  $L = F / e$
  OR
  $e = F / L$; [1 mark]

[Total: 1 mark]

- This is known as Faraday's Law which defines one faraday as the amount of electric charge carried by 1 mole of electrons or 1 mole of singly charged ions
- You need to know and be able to apply this relationship

2d

d) The value of the Avogadro constant is:

- $L = 9.65 \times 10^4 \div 1.60 \times 10^{-19} = 6.02 \times 10^{23}$; [1 mark]

[Total: 1 mark]

- Make sure that the value that you obtain looks right - students sometimes make mistakes entering values into their calculator and write down whatever value is shown
- Even if you can't remember the exact value of Avogardo's constant, you should

be aware that it is a very large number so if you get a small value or a number with a negative power - check your calculation

3a

a)

i) The substance used for the electrode in a standard hydrogen electrode is:

- Platinum; [1 mark]

ii) Platinum is used as the electrode because

- It is inert
  AND
  It conducts electricity; [1 mark]
- It acts as a source of electrons; [1 mark]

iii) The standard conditions used in a standard hydrogen electrode are:

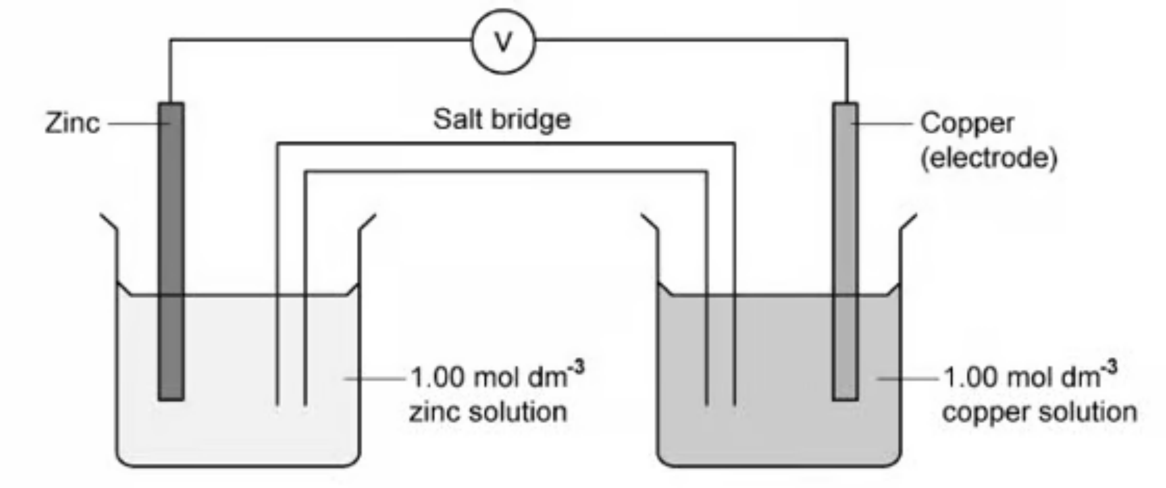
- 1.00 mol dm^{-3} (H^+ / hydrogen ions / concentration); [1 mark]
- 298 K; [1 mark]
- 101 kPa; [1 mark]

[Total: 6 marks]

- Platinum provides the electrons to allow the hydrogen molecules to reach equilibrium with the hydrogen ions:
  - $2H^+ (aq) + 2e^- \rightleftharpoons H_2 (g)$
- You must make sure that you know the exact reaction conditions
  - You could be asked to state these, as in this question
  - You could also be asked to spot something wrong with a reaction diagram

3b

b)
i) The completed electrochemical cell is:



- Correctly labelled electrode; [1 mark]
- Copper solution / Cu^{2+} solution / Cu^{2+} (aq) labelled
  AND
  1 mol dm^{-3}; [1 mark]
- Completed
  AND
  Labelled salt bridge; [1 mark]
- Voltmeter connected between the two electrodes; [1 mark]

ii) The half-equations are:

- $Zn^{2+}(aq) + 2e^{-} \rightarrow Zn(s)$; [1 mark]
- $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$; [1 mark]

[Total: 6 marks]

- To complete the cell diagram a salt bridge and voltmeter must be added and the correct concentration must be given
- The salt bridge must be drawn so that it is clear that it is submerged in the copper solution otherwise the ions will not be able to flow
- A voltmeter must be added so that the EMF of the cell can be measured
- Standard conditions state that the solutions should have a concentration of 1.00 mol dm^{-3}
- Half-equations using the IUPAC convention must be written as **reduction** reactions and include state symbols

3c

c)

i) The half-equation for the reaction that occurs at the positive electrode is:

- $Ag^+ + e^- \rightarrow Ag$; [1 mark]

ii) The half-equation for the reaction that occurs at the negative electrode is:

- $Cu \rightarrow Cu^{2+} + 2e^-$; [1 mark]

iii) The overall equation for the cell is:

- $Cu\,(s) + 2Ag^+\,(aq) \rightarrow Cu^{2+}\,(aq) + 2Ag\,(s)$; [1 mark]

[Total: 3 marks]

- The equation at the positive electrode is the right-hand side of the cell representation which is the one with the more positive electrode potential
- The equation at the negative electrode is the left-hand side of the cell representation which is the one with the more negative electrode potential
- To complete the overall equation ensure that there is the same number of electrons in both half equations (by multiplying one or both half equations until they are equal)
- The electrons on each side of the equation should then balance out and so are removed from the equation

3d

d)

i) The salt bridge provides an electrical connection between the two solutions because:

- It has mobile / free ions
  OR
  It has ions that can move through it
  OR
  The ions (in the salt bridge) are not fixed in position; [1 mark]

ii) Potassium chloride would **not** be suitable for use in the salt bridge of this cell because:

- Chloride ions / Cl^- ions would react with / form a precipitate with the silver ions / Ag^+ ions
  OR
  AgCl solid would form; [1 mark]

[Total: 2 marks]

- A salt bridge is used to keep the two half-reactions separate but provide an ionic connection between them so that the ions can move and the circuit is complete
- The ionic salt solution used in the salt bridge must be inert
  - Potassium nitrate is commonly used

4a

a) The material that should be used for each electrode is:

- Electrode 1: Platinum; [1 mark]
- Electrode 2: Platinum; [1 mark]

[Total: 2 marks]

- The half-cells both contain ions that are in different oxidation states
  - This means that inert platinum electrodes will be needed

4b

b) Solution **1** could be:

- 1 mol dm^{-3} HCl / HNO_3; [1 mark]

[Total: 1 mark]

- As it is a standard hydrogen electrode, the hydrogen ion concentration in the

AgCl solid would form; [1 mark]

[Total: 2 marks]

- A salt bridge is used to keep the two half-reactions separate but provide an ionic connection between them so that the ions can move and the circuit is complete
- The ionic salt solution used in the salt bridge must be inert
  - Potassium nitrate is commonly used

4a

a) The material that should be used for each electrode is:

- Electrode 1: Platinum; [1 mark]
- Electrode 2: Platinum; [1 mark]

[Total: 2 marks]

- The half-cells both contain ions that are in different oxidation states
  - This means that inert platinum electrodes will be needed

4b

b) Solution **1** could be:

- 1 mol dm^{-3} HCl / HNO_3; [1 mark]

[Total: 1 mark]

- As it is a standard hydrogen electrode, the hydrogen ion concentration in the solution must be 1.0 mol dm^{-3}
- HCl is typically used as it is a strong, monobasic acid that will fully ionise, so 1 mol dm^{-3} HCl will provide 1 mol dm^{-3} H^+ ions
- **Careful**: You cannot suggest a weak acid, such as 1.0 mol dm^{-3} of ethanoic acid, as it will not fully ionise, therefore the solution will not contain 1 mol dm^{-3} H^+ ions
  - A dibasic strong acid is also not suitable, for example, 0.5 mol dm^{-3} H_2SO_4 which would contain 1 mol dm^{-3} H^+ ions if fully ionised, however, some HSO_4^- ions will remain present in the solution and not fully ionise
  - Similarly, 0.33 mol dm^{-3} of H_3PO_4 would not be suitable

4c

c) The mass of $Cr_2(SO_4)_3 \cdot 18H_2O$ should also be used is:

- Moles of $K_2Cr_2O_7 = \frac{14.71}{294.2} = 0.050$ (mol); [1 mark]
- Moles of $Cr_2(SO_4)_3 \cdot 18H_2O = \frac{0.050}{2} = 0.025$ (mol); [1 mark]
- Mass of $Cr_2(SO_4)_3 \cdot 18H_2O = 0.025 \times 716.3 = 17.91$ (g); [1 mark]

[Total: 3 marks]

- In a standard solution, there should be 1 mol dm^{-3} of both $Cr_2O_7^{2-}$ and Cr^{3+}
- This means that there should also be 0.05 moles of Cr^{3+} (as the volume is the same)
- As 1 mole of $Cr_2(SO_4)_3 \cdot 18H_2O$ contains 2 moles of Cr^{3+}, the number of moles of $Cr_2(SO_4)_3 \cdot 18H_2O$ that will contain 0.05 moles of Cr^{3+} is 0.05/2 = 0.025 moles

4d

d) Solution **2** is best acidified with H_2SO_4 instead of HCl or HBr because:

- Potassium dichromate / the dichromate ion is strongly oxidising / an oxidising agent; [1 mark]
- Chloride ions / Cl^- and bromide ions / Br^- are more easily oxidised than sulfate ions / SO_4^{2-}; [1 mark]

[Total: 2 marks]

- You should be familiar with the role of potassium dichromate as an oxidising agent from its use to oxidise alcohols
- The halide ions are easily oxidised, causing them to lose electrons:
  - $2Cl^- \rightarrow Cl_2 + 2e^-$
- If sulfuric acid is the preferred acid to use, this implies that the sulfate ion is less easily oxidised
- In fact, potassium dichromate is not capable of oxidising sulfate ions
  - So, sulfuric acid is always used to acidify dichromate solutions.

5a

a)

i) A half-equation including state symbols for the reaction of copper at the anode is:

- $Cu (s) \rightarrow Cu^{2+} (aq) + 2e^{-}$; [1 mark]

ii) Silver remains as the metal because:

- E^{θ} for Ag^{+} / Ag is +0.80 V which is more positive than +0.34 V for Cu^{2+} / Cu; [1 mark]
- So, it is less easily oxidised; [1 mark]

iii) Predicting what happens to the nickel at the anode:

- E^{θ} for Ni^{2+} is -0.25V; [1 mark]
- Ni is readily oxidised and goes into solution as Ni^{2+} (aq); [1 mark]

iv) The half-equation including state symbols for the main reaction at the cathode is:

- $Cu^{2+} (aq) + 2e^{-} \rightarrow Cu (s)$; [1 mark]

v) Zinc is not deposited on the cathode because:

- E^{θ} for Zn^{2+} / Zn is negative, so Zn^{2+} is not easily reduced; [1 mark]

[Total: 7 marks]

- You are told in the question that the anode is made from impure copper and that copper goes into solution at the anode, so that means Cu^{2+} (aq) is formed from Cu
  - It does this by losing electrons giving the half-equation
    - $Cu (s) \rightarrow Cu^{2+} (aq) + 2e^{-}$
  - You would not gain this mark if you didn't write the state symbols
- The reverse of this reaction occurs at the cathode
- **Remember**:
  - The more positive the value of E^{θ}, the more easily the species on the left of the electrode reaction is reduced
  - The more negative the value of E^{θ}, the more easily the species on the right of the electrode reaction is oxidised

5b

b) The blue colour fades because:

- Cu^{2+} is replaced by Zn^{2+} / Ni^{2+}

OR

The concentration of Cu^{2+} decreases; [1 mark]

[Total: 1 mark]

- The blue colour associated with $CuSO_4$ (aq) is due to Cu^{2+} ions
- As the electrolysis proceeds, the Cu^{2+} ions in solution are gaining electrons to form copper which is deposited on the cathode
- Some of these Cu^{2+} ions are replaced as the impure copper anode releases Cu^{2+} ions
- However, because the Ni and Zn impurities in the copper anode are both oxidised, some of the Cu^{2+} ions are replaced with Ni^{2+} and Zn^{2+} ions
- This means that the concentration of Cu^{2+} will decrease over time, and the blue colour will gradually fade

5c

c)

i) The number of moles of copper produced at the cathode is:

- Mole of Cu = 225/63.5 = 3.54(3) (mol); [1 mark]

ii) The number of moles of electrons needed to produce this copper is:

- Moles of electrons = 2 x 3.54 = 7.09 (7.087) (mol); [1 mark]

iii) The number of moles of electrons that passed through the cell are:

- Number of coulombs, $Q = It$ = 20 x 10 x 60 x 60 = 7.2 x 10^5 C; [1 mark]
- Moles of electrons = $\frac{7.2 \times 10^5}{9.65 \times 10^4}$ = 7.46(1) (mol); [1 mark]

iv) The percentage of the current through the cell that has been 'wasted' in dissolving the impurities at the anode is:

- $\frac{(7.461 - 7.087)}{7.461} \times 100 = 5.01\%$; [1 mark]

[Total: 5 marks]

- A final value of between 4.98 to 5.10% is accepted as the value will be slightly different if you round your answer at each stage compared with if you use the calculator value throughout

- **Remember**: When using $Q = It$, time must be in seconds
  - To convert from hours to seconds, multiply by 60 x 60 or 3600
- The Faraday constant, F, is the amount of charge carried by 1 mole of electrons
- To calculate the number of moles of electrons, divide the calculated charge, Q, by the Faraday constant
- To find the percentage of current wasted, you need to find the number of moles of electrons that weren't used to produce 225 g of copper = 7.461 - 7.087 = 0.374 moles
- Then find what this is as a percentage of the total number of moles of electrons that passed through the cell
  - $\frac{0.374}{7.461}$ x 100 = 5.01%

5d

d) To explain what would happen to iron during this process:

- Fe^{2+} / Fe E^{θ} is more negative than Ni^{2+} / Ni; [1 mark]
- So iron will dissolve; [1 mark]

[Total: 2 marks]

- As the electrode potential of Fe^{2+} / Fe is more negative, Fe is more easily oxidised so it will lose electrons:
  - $Fe (s) \rightarrow Fe^{2+} (aq) + 2e^{-}$
- This means that the iron will dissolve into the solution